Status	Finished
Started	Sunday, 4 October 2026, 7:16 PM
Completed	Sunday, 4 October 2026, 7:21 PM
Duration	4 mins 44 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

- a. The search continues in the left or right sublist ✓
- b. The search continues from the beginning
- c. The search stops
- d. The list is sorted

Question **2**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

- a. Hash search
- b. Binary search ✓
- c. Linear search
- d. Sequential search

Question **3**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- a. When the list is sorted
- b. When the list contains duplicate elements
- c. When the search algorithm finishes
- d. When the element is found in the list ✓

Question **4**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By using a hash function
b. By dividing the list length by two ✓
c. By comparing each element sequentially
d. By starting from the first element

Question **5**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the right sublist ✓
b. In the middle of the list
c. At the beginning of the list
d. In the left sublist

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It can only be applied to large lists
b. It requires the list to be sorted ✓
c. It does not work with negative numbers
d. It is slower than linear search for small lists

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is sorted ✓
b. When the list is unsorted
c. When the list contains duplicate elements
d. When the list is very small

Question **8**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Finding
b. Mining
c. Discovering
d. Searching ✓

Question **9**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O🥐
b. O(n log n)
c. O(log n) ✓
d. O(1)

Question **10**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. log n
b. n ✓
c. 1
d. n/2

Question **11**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Selection

b. Insertion

c. Bubble ✓

d. Complexity

Question **12**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Both (1) & (3)

b. Linear

c. Hash

d. Binary ✓

Question **13**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 1000

b. 100

c. 10 ✓

d. 500

Question **14**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 90 and 100

b. 94 and 99

c. 90 and 99 ✓

d. 89 and 94

Question **15**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the entire list

b. At the beginning of the list

c. In the left sublist ✓

d. In the right sublist